In this approach, then, the abrupt collapse of a wavefunction is completely unsurprising: it is nothing more than the abrupt change in knowledge that we all experience when we learn something new.

A second approach, initiated in 1957 by Wheeler's student Hugh Everett, denies that wavefunctions ever collapse. Instead, each and every potential outcome embodied in a wavefunction sees the light of day; the daylight each sees, however, streams through its own separate universe. In this approach, the *Many Worlds interpretation,* the concept of "the universe" is enlarged to include innumerable "parallel universes"—innumerable versions of our universe—so that anything that quantum mechanics predicts *could* happen, even if only with minuscule probability, *does* happen in at least one of the copies. If a wavefunction says that an electron can be here, there, and way over there, then in one universe a version of you will find it here; in another universe, another copy of you will find it there; and in a third universe, yet another you will find the electron way over there. The sequence of observations that we each make from one second to the next thus reflects the reality taking place in but one part of this gargantuan, infinite network of universes, each one populated by copies of you and me and everyone else who is still alive in a universe in which certain observations have yielded certain outcomes. In one such universe you are now reading these words, in another you've taken a break to surf the Web, in yet another you're anxiously awaiting the curtain to rise for your Broadway debut. It's as though there isn't a single spacetime block as depicted in Figure 5.1, but an infinite number, with each realizing one possible course of events. In the Many Worlds approach, then, no potential outcome remains merely a potential. Wavefunctions don't collapse. Every potential outcome comes out in one of the parallel universes.

A third proposal, developed in the 1950s by David Bohm—the same physicist we encountered in Chapter 4 when discussing the Einstein-Podolsky-Rosen paradox—takes a completely different approach.[8] Bohm argued that particles such as electrons *do* possess definite positions and definite velocities, just as in classical physics, and just as Einstein had hoped. But, in keeping with the uncertainty principle, these features are hidden from view; they are examples of the *hidden variables* mentioned in Chapter 4. You can't determine both simultaneously. For Bohm, such uncertainty represented a limit on what we can know, but implied